

Newton's law of gravitation

Newton's law of gravitation $\implies F \propto \frac{m_1 \times m_2}{r^2}$: r is the distance between two point masses and where F is an attractive force ($F < 0$ N).

For an object on Earth's surface, the gravitational force is:

$$F = \frac{Gm \times m_E}{r_E^2}$$
$$\text{Weight} = mg$$
$$mg = \frac{Gm \times m_E}{r_E^2}$$
$$\therefore g = \frac{G \times m_E}{r_E^2}$$

g is the gravitational field strength.

Not exclusive to Earth and can be used anywhere in a gravitational field,

$$g = \frac{\text{force acting on a small test object}}{\text{mass of the test object}} = \frac{F}{m}$$

g has two units:

N kg^{-1} when dealing with gravitational field strength

ms^{-1} when dealing with acceleration of free fall

Test objects are required because objects can contribute to the gravitational field where it is placed. Test objects would have negligible effect on the original gravitational field it is placed in since mass of test object $\ll$ mass of object producing the field.